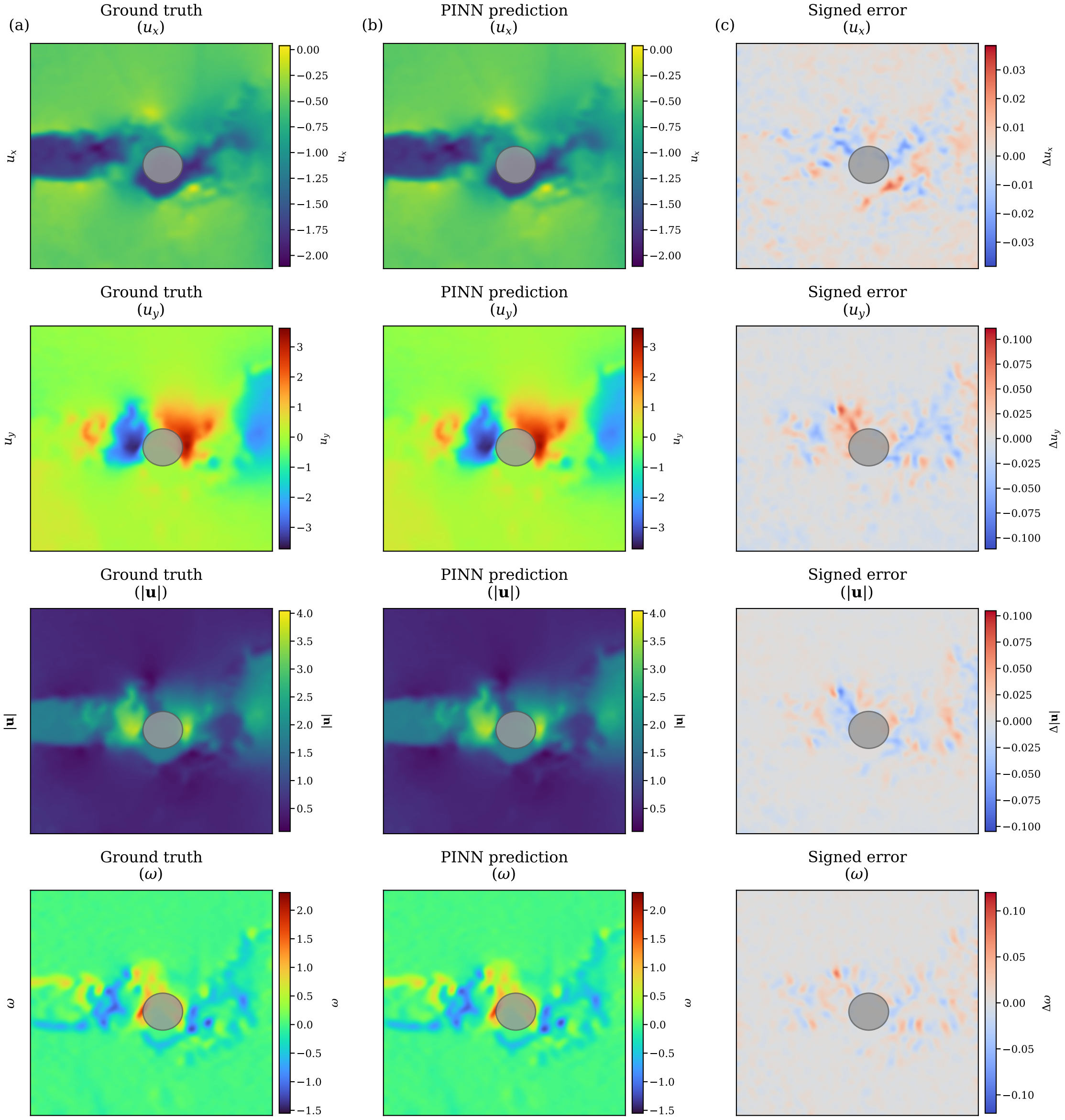
(a)
Ground truth
(u_x)
u_x
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
(b)
PINN prediction
(u_x)
(c)
Signed error
(u_x)
Δu_x
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
Ground truth
(u_y)
u_y
3
2
1
0
−1
−2
−3
PINN prediction
(u_y)
Signed error
(u_y)
Δu_y
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Ground truth
($|\mathbf{u}|$)
$|\mathbf{u}|$
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
PINN prediction
($|\mathbf{u}|$)
Signed error
($|\mathbf{u}|$)
$\Delta|\mathbf{u}|$
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Ground truth
(ω)
ω
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
PINN prediction
(ω)
Signed error
(ω)
$\Delta\omega$
0.10
0.05
0.00
−0.05
−0.10